How Much Does Limit Order Book Microstructure Add Beyond Quote Geometry?

Evidence from 70 days of cryptocurrency data at
retail-accessible sampling frequency

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Abstract

A trader deciding where to post a limit order knows three things for free: which side they intend to quote, how far from mid, and how long they are willing to wait. Whether limit order book (LOB) microstructure adds anything beyond those three choices is an empirical question, and one rarely posed against a baseline simple enough to make the answer interpretable.

We pose it directly. Using 269,330 LOB snapshots collected over a 70.5-day calendar window from Kraken across BTC/USD, ETH/USD and SOL/USD, we predict whether the opposite best quote reaches or crosses a limit price posted at a given distance from mid, within a fixed horizon, and we benchmark against a 16-cell lookup table containing only the mean historical crossing rate for each combination of quote offset, horizon and side. That table is a demanding reference: both conditioning variables carry large monotone effects on the crossing rate, spanning 0.46 across offsets on BTC/USD at 300 s and comparable ranges elsewhere (Table 2).

Microstructure features beat it, modestly and consistently. Under a snapshot-grouped temporal split with a 990-second embargo, gradient-boosted trees over 63 microstructure features gain +0.048 and +0.060 AUC on BTC and ETH (Table 3 reports all figures to four decimals). Under a blocked split constructed so that training and test partitions span overlapping price regimes, the gain is positive on all three assets. Rotating the arbitrary block phase gives ten phase assignments with disjoint test sets per asset, and all thirty are positive: means of +0.047, +0.038 and +0.035, with no rotation below +0.013. Across-phase standard deviations are 0.0052, 0.0144 and 0.0103.

The split scheme matters, and revealed something we would otherwise have misreported. Under the temporal split, training and test partitions occupied almost disjoint price ranges: overlap of 0.129, 0.073 and 0.251. SOL/USD appeared to show a *negative* microstructure effect there, which reversed to +0.027 once the partitions spanned comparable price regimes. We report both schemes throughout, since neither substitutes for the other: the temporal split respects forecasting order, and the blocked split reduces temporal distribution shift between partitions.

We also compare model classes, holding the input representation fixed. Gradient-boosted trees lead an LSTM on BTC and ETH under both representations, but on SOL/USD the ordering flips with representation: trees lead by 0.0013 on 109 features and trail by 0.0475 on the 63-feature microstructure subset. We draw no general ordering. An earlier version claimed model class was decisive; that rested on a defective survival head.

Finally, we document five defects in our own pipeline, each of which produced plausible, publishable-looking numbers, and the specific checks that caught them. We consider this the most transferable part of the work.

Keywords: limit order book, fill probability, crossing proxy, market microstructure, gradient boosting, baselines, reproducibility

JEL: G12, G14, C45, C61

1 Introduction

Consider a trader posting a limit order. Three quantities are entirely under their control and known before any model is consulted: which side they quote, the distance from mid at which they quote, and how long they will leave the order resting. All three bear on whether the order is reached. Quoting closer to mid is reached more often; waiting longer is reached more often.

The question this paper asks is what remains. Given those three choices, does the state of the order book — spreads, imbalances, depth, recent volatility — carry additional information about whether the order will fill?

The question is worth posing carefully because the obvious answer, that of course microstructure is informative, is not the same as a measured quantity. A large literature reports LOB models achieving substantial predictive accuracy [Zhang et al., 2019, Maglaras et al., 2022, Arroyo et al., 2024], but reported accuracy conflates what the model learned with what was mechanically available from quote placement. A model can post a respectable AUC while adding nothing a lookup table does not already provide.

1.1 What we find

Microstructure adds a modest, consistent amount. Under a blocked split that reduces temporal distribution shift between partitions, gradient-boosted trees over microstructure features gain +0.048, +0.049 and +0.027 AUC over the lookup reference on BTC/USD, ETH/USD and SOL/USD. Rotating the block phase yields ten phase assignments with disjoint test sets per asset, and all thirty gaps are positive, with means of +0.047, +0.038 and +0.035 and no rotation below +0.013.

The finding is robust to split scheme on two of three assets and changes sign on the third. We examine this split sensitivity in Section 6.

1.2 What we no longer claim

Version 3 of this work argued that model class was decisive: that trees recovered signal a deep survival model could not. That claim rested on a survival decoder being fed binary targets with a constant observed time, which pinned its training loss at 16.1181 and returned AUC of exactly 0.5000 on every arm. With a binary classification head the LSTM trains normally, and the ordering becomes representation-dependent. Holding inputs fixed, trees lead on BTC and ETH under both representations; on SOL they lead by 0.0013 with all 109 features but trail by 0.0475 on the 63-feature microstructure subset. We therefore draw no general model-class ordering.

Versions 1 and 2 framed the work around execution latency, proposing that prediction could substitute for speed for traders operating at 200 ms–2 s delays. That framing does not survive: the labels supporting it were generated on a hardcoded 100 ms synthetic clock and, measured against real timestamps, the "2000 ms" horizon was a 511-second median window, an overstatement of

256×. The horizons we can actually resolve on this feed are 300 and 900 seconds, which is a different problem and is the one this paper addresses.

1.3 Contributions

1. A measurement of the marginal value of LOB microstructure over quote geometry, against an explicit trivial reference, on three assets under two split schemes (Sections 5, 6).
2. Evidence that the sign of a microstructure result can depend on split scheme under substantial temporal distribution shift, with the diagnostic that detects it and a phase-rotation test that quantifies dispersion across block-phase assignments (Section 6).
3. A dataset of 269,330 timestamped 10-level LOB snapshots, with the full collection, labelling and evaluation pipeline released publicly (Section 3).
4. A catalogue of five pipeline defects that produced plausible results, and the checks that caught each (Section 7).

1.4 Scope

Observed under the forecasting-ordered split. In our Kraken sample, three cryptocurrency pairs from June to August 2026, microstructure features carry predictive content for our replay-defined crossing outcome beyond side, quote offset and horizon on BTC/USD and ETH/USD, at +0.048 and +0.060 AUC, recoverable by gradient-boosted trees. SOL/USD is negative under this split.

Observed under the blocked-split diagnostic. The same gain appears on all three assets across all ten block-phase rotations, 30 of 30 positive, with means of +0.047, +0.038 and +0.035. That split was introduced after inspecting the temporal results (Section 6) and remains an information-content diagnostic rather than a forecasting evaluation.

Not established: economic exploitability. AUC is not profit. We run no backtest, model no transaction costs, and make no claim that this edge survives the spread, maker fees, queue position or adverse selection.

Not established: realised fills. Labels are reconstructed from subsequent book snapshots and use no trade or execution data. A quote is recorded as crossed when the opposite best quote reaches or crosses its price, which is a proxy in two directions. It is optimistic for an order arriving late in the queue, since a crossing does not establish that the resting order executed. It is also potentially a false negative: with median gaps of 11.5 to 48.0 seconds, a crossing that occurs and reverses entirely between two observed snapshots is not recorded.

Not established: calibration. AUC measures ranking, not whether a predicted 0.70 corresponds to a 70% crossing rate.

Not established: forecasting performance. Our blocked-split results train on data from after the test period. They measure whether information is present, not whether it could have been traded in real time. The temporal results address that and are reported alongside.

Not established: generality. One venue, one 70-day window, three cryptocurrency pairs, one feed type, two horizons.

2 Related Work

2.1 Fill probability estimation

Lo et al. [2002] established econometric foundations for execution probability modelling. Maglaras et al. [2022] introduced a deep learning approach treating execution as a time-to-fill distribution. Arroyo et al. [2024] advanced this with a convolutional-transformer using survival analysis, which handles the right-censoring inherent in limit order data.

That work uses tick-level message data. Ours is snapshot data at median gaps of 11.5 to 48.0 seconds, three to four orders of magnitude coarser, so our results are not directly comparable and should not be read as contradicting theirs. What our setting permits, which theirs does not emphasise, is a clean separation between the mechanical contribution of quote placement and the informational contribution of book state.

2.2 Baselines in financial machine learning

Our methodological point — that a trivial reference is necessary to interpret a reported accuracy — is not novel in general, but is inconsistently applied in this literature. López de Prado [2018] discusses at length the ways financial backtests overstate performance, including through improper cross-validation on serially correlated data, and proposes purging and embargoing, which we adopt in Section 4.4.

2.3 Tabular deep learning

Systematic evaluations report gradient-boosted trees remaining competitive with neural networks on tabular data, particularly at moderate sample sizes with heterogeneous features [Grinsztajn et al., 2022, Shwartz-Ziv and Armon, 2022]. Our features are engineered per-snapshot scalars, which is tabular in structure. Our result is directionally consistent with that literature but weaker than we previously claimed: trees lead on BTC and ETH under both representations, while the ordering on SOL depends on the representation.

2.4 Market making theory

The framing that motivated earlier versions of this work derives from Avellaneda and Stoikov [2008], with latency treated as a structural risk factor by Cartea and Sánchez-Betancourt [2023] and Lehalle and Mounjid [2022]. We retain none of that framing here, because the horizons our data resolve are 300 and 900 seconds rather than the sub-second regime those models address. We note the connection only to be explicit that this paper does not extend it.

3 Data

3.1 Collection

LOB data were collected from Kraken via WebSocket API v2, subscribing to the `book` channel at depth 10 for BTC/USD, ETH/USD and SOL/USD. Updates are timestamped at receipt and stored with the full 10-level ladder in Parquet.

Collection ran from 15 June to 24 August 2026, a span of 70.5 days. The collector runs under a systemd unit with `Restart=always` and restarted automatically three times after WebSocket disconnections. An earlier configuration using a detached terminal session and a bounded reconnect loop failed silently on 20 July and stayed down for fourteen days.

Table 1: Collected data. Gaps are between consecutive book updates; the feed is event-driven, so sampling is irregular. Mean gaps include the 14-day collector outage described above and therefore approximate calendar span divided by snapshot count; the median is the better measure of normal feed cadence.

Asset	Snapshots	Tick	Median gap	Mean gap*	Label rows
BTC/USD	111,002	0.10	11.5 s	54.9 s	1,760,728
ETH/USD	34,380	0.01	48.0 s	177.2 s	515,576
SOL/USD	123,948	0.01	26.2 s	49.1 s	1,971,128
Total	269,330				4,247,432

*Includes the collector outage; see caption.

3.2 Labels

For each snapshot we simulate limit orders at offsets of 1, 2, 5 and 10 basis points from mid on each side, rounded to the venue tick so the quote is postable, and record whether the opposite best quote reached or crossed that level within horizons of 300 and 900 seconds. A buy at price p is recorded as crossed if min(best ask) $\leq p$ within the horizon; a sell if max(best bid) $\geq p$.

We call this a *replay-defined crossing*. It uses no trade or execution data, only subsequent book snapshots, so it does not establish that a resting order executed.

Three choices here are deliberate and were wrong in earlier versions.

Horizons are resolved against real timestamps, not a fixed snapshot count. Earlier versions applied a hardcoded 100 ms interval inherited from a synthetic data generator (Section 7).

Offsets are in basis points, not ticks. Venue ticks are not economically comparable across these assets: BTC's 0.10 tick on a ∼65,000 price is 0.00015% of mid, while SOL's 0.01 tick on ∼150 is 0.0067%, roughly 44× larger. Under tick offsets the 1–5 range spanned a 0.176 crossing-rate gradient on SOL but only 0.011 on BTC, so "offset" meant different things per asset.

Unobservable rows are dropped, not labelled as non-crossings. On a feed with a median gap of 11.5 to 48.0 seconds, some snapshots have no subsequent observation inside a horizon. Recording those as "did not cross" manufactures the label. Observability is 88.7% at worst (ETH at 300 s) and 98.8% or above at 900 s. This is why label counts are not snapshots × 16.

Table 2: Crossing rate by horizon and quote offset. Both conditioning variables carry strong monotone structure, which is what makes the lookup reference demanding.

Asset	Horizon	1 bps	2 bps	5 bps	10 bps
BTC/USD	300 s	0.697	0.622	0.433	0.237
	900 s	0.815	0.766	0.626	0.440
ETH/USD	300 s	0.603	0.540	0.398	0.245
	900 s	0.740	0.696	0.582	0.432
SOL/USD	300 s	0.577	0.517	0.387	0.242
	900 s	0.753	0.712	0.612	0.466

3.3 Features

From each snapshot we construct 109 features: raw price and volume levels, normalised prices and volumes, mid-price, spread, spread in basis points, order book imbalance, depth imbalance, and rolling statistics and lagged differences.

Two subsets are used, both defined by keyword match on feature names. The *microstructure* subset holds the 63 features matching spread, imbalance, order book imbalance, depth, volume or volatility terms and not containing `price`. The *price-named* subset holds the 41 features whose names contain `price`, raw or normalised. We use that label rather than "price levels" because the group includes mid-normalised columns as well as raw ones.

The two do not partition the feature set. Five features fall into neither: `return_1` and `return_lag_`k for $k \in \{1, 2, 3, 5\}$. These are lagged mid-price returns, so they are price-derived in substance, but the keyword rule matches the literal string `price` and does not capture them. The price-named subset therefore contains 41 raw or normalised price columns; the five lagged returns enter only the full 109-feature arm.

We state this because $109 - 63 = 46$ counts non-microstructure columns rather than price columns, and the two differ by exactly these five returns. The microstructure subset excludes both groups: returns match no microstructure keyword either. What those 63 features exclude is direct price-level columns and directional lagged returns. They are not free of price information in a strict sense, since spreads and realised volatility are themselves functions of prices; what they exclude is the price *level* and the *direction* of recent moves. Headline results use that subset, for reasons given in Section 6.

4 Method

4.1 The reference

Our benchmark is a lookup table: the mean crossing rate for each (offset, horizon, side) cell, 16 cells, estimated on training data and applied unchanged to test data. It has no learned parameters and no access to the order book. Table 2 shows what it captures.

4.2 Models

Gradient-boosted trees (XGBoost, 400 estimators, depth 6, learning rate 0.05, subsample 0.8) on the current snapshot's features plus the three conditioning variables. No temporal window.

Binary LSTM: two layers, hidden dimension 64, over a 50-snapshot window, with the conditioning vector concatenated before a single logit head trained with binary cross-entropy. The two models receive different inputs: the LSTM sees a 50-snapshot history, the GBT only the current snapshot. We do not treat longer history as automatically advantageous, particularly on a sparse and irregular sequence, and state the asymmetry rather than characterising it.

4.3 Sampling

Model fits do not use all 4,247,432 label rows. For each asset we draw a uniform random sample without replacement of 200,000 rows, using a fixed seed (0), and restore temporal order before splitting. The cap bounds LSTM training time, and the same sampled set is used by every arm so the comparison is on identical rows.

Sampling occurs before the split, so partition proportions are computed on the sampled rows. This is 11.4% of available BTC/USD labels, 38.8% of ETH/USD and 10.1% of SOL/USD, so the effective sample is more nearly balanced across assets than Table 1 suggests. Because sampling is uniform over rows rather than stratified by snapshot, it also thins the number of label rows per snapshot; the grouped split (below) still operates on whole snapshots.

4.4 Splitting and embargo

Each snapshot generates up to 16 label rows sharing identical features, so a row-wise split permits a snapshot to appear in both training and test data. All splits are therefore *snapshot-grouped*.

Labels look up to 900 seconds forward. Every partition boundary carries a 990-second embargo, following the purging logic of López de Prado [2018], so a training label's forward window cannot reach into test data.

We report two schemes. The **temporal** split takes the first 70% of snapshots for training, the next 15% for validation and the last 15% for test, which respects forecasting order. The **blocked** split cuts the timeline into 150 contiguous blocks assigned in a repeating pattern, so all partitions span the full price history. Section 6 explains why both are necessary.

4.5 Metric

We report crossing AUC: area under the ROC curve for the binary crossed/not-crossed task, computed exactly via the Mann-Whitney rank-sum identity with ties credited 0.5. It measures discrimination. It does not assess calibration, so these results do not establish that predicted probabilities are usable as probabilities.

5 Results

5.1 Main comparison

Table 3: Crossing AUC under the temporal split, three seeds. Each cell gives the mean, the seed standard deviation, and the gap over the lookup reference for that asset. GBT sees one snapshot; LSTM sees fifty.

Asset	Lookup	GBT micro (63)	GBT all (109)	LSTM micro (63)	LSTM all (109)
BTC/USD	0.6828	0.7307 ±0.0006 +0.0479	0.7262 ±0.0012 +0.0434	0.6857 ±0.0021 +0.0029	0.6903 ±0.0043 +0.0075
ETH/USD	0.6597	0.7193 ±0.0013 +0.0596	0.6908 ±0.0010 +0.0311	0.6983 ±0.0046 +0.0386	0.6669 ±0.0036 +0.0072
SOL/USD	0.6673	0.6568 ±0.0035 −0.0105	0.7056 ±0.0010 +0.0383	0.7043 ±0.0046 +0.0370	0.7043 ±0.0074 +0.0370

Seed standard deviations, shown beneath each mean, span 0.0006–0.0035 for GBT arms and 0.0021–0.0074 for LSTM arms, so the differences discussed below exceed run-to-run variation. They quantify model-initialisation variance only, not variance over data partitions.

Two observations, and one anomaly deferred to Section 6.

Microstructure features add measurable signal on BTC and ETH: +0.048 and +0.060 for the tree model on the 63-feature subset. SOL shows −0.011, the only negative entry in the table, which we resolve below.

Adding the 46 non-microstructure columns (41 price-named and five lagged returns) *hurts* on BTC and ETH, with GBT-all below GBT-micro by 0.005 and 0.028. On SOL they appear to help substantially. This inconsistency is the same anomaly.

5.2 Model class

Comparing model classes requires holding the input representation fixed. An earlier version of this paper compared the better-performing arm of each model class, selected by test AUC. That is post-hoc selection on the test set: it allows the representation to differ by model and by asset according to which number is larger, and it produced a claim that trees lead on all three assets. Table 4 instead reports both representations separately.

Table 4: GBT minus LSTM, holding the input representation fixed. Positive favours trees. Arms are not selected on test performance.

	Microstructure (63)		All (109)	
Asset	GBT − LSTM	Leader	GBT − LSTM	Leader
BTC/USD	+0.0450	GBT	+0.0359	GBT
ETH/USD	+0.0210	GBT	+0.0239	GBT
SOL/USD	−0.0475	LSTM	+0.0013	GBT

The ordering is representation-dependent. On the 109-feature representation trees lead on all three assets, though by only 0.0013 on SOL/USD, which is within the seed standard deviation of the SOL LSTM arm (0.0074). On the 63-feature microstructure representation trees lead on BTC and ETH but trail on SOL by 0.0475.

We therefore do not claim a general ordering between model classes on this data. Trees are ahead in four of the six comparisons, clearly so on BTC under both representations, and the one reversal is substantial. Version 3 of this work reported the LSTM at exactly 0.5000 on every arm and concluded model class was decisive; that was a defective survival head rather than a result (Section 7). The corrected comparison supports no such conclusion.

One arm warrants caution. LSTM-all on ETH moved its training loss by only 0.0078, against 0.1593 for LSTM-micro on the same data. That arm is likely undertrained rather than genuinely uninformative.

6 Split-Scheme Sensitivity

6.1 Diagnosis

SOL's negative microstructure gap prompted a check on how the temporal split partitions price levels. The result applies to all three assets.

Over a trending 70-day window, a temporal split leaves training and test partitions occupying nearly disjoint price ranges. ETH trains on $1510–1944 and tests on $1869–2545, an overlap of 0.073. A tree splitting on absolute price can identify the partition almost perfectly, which makes any gain attributable to price-derived features difficult to interpret.

Table 5: Overlap between training and test price ranges, by split scheme. Low overlap means absolute price nearly identifies the partition.

Asset	Temporal	Blocked
BTC/USD	0.129	0.815
ETH/USD	0.073	0.855
SOL/USD	0.251	0.745

The blocked split cuts the timeline into 150 contiguous blocks assigned in a repeating pattern with the same 990-second embargo at every boundary, raising overlap to 0.745–0.855.

6.2 Effect

Table 6: Gap over lookup by feature group and split scheme, three seeds. For the blocked microstructure arm the underlying AUCs are 0.7519, 0.7141 and 0.6955 against lookup references of 0.7040, 0.6650 and 0.6681, with seed standard deviations of 0.0004, 0.0006 and 0.0003.

	Microstructure (63)		Price-named (41)		All (109)	
Asset	Temporal	Blocked	Temporal	Blocked	Temporal	Blocked
BTC/USD	+0.0479	+0.0479	+0.0384	+0.0383	+0.0434	+0.0456
ETH/USD	+0.0596	+0.0491	−0.0001	+0.0370	+0.0312	+0.0513
SOL/USD	−0.0105	+0.0274	+0.0312	+0.0115	+0.0383	+0.0295

Three findings.

The microstructure result is robust where it was already positive. BTC is identical to four decimals (+0.0479 under both); ETH holds within 0.011.

SOL's negative result is sensitive to the split scheme. It moves from −0.0105 under the forecasting-ordered split to +0.0274 once partitions span comparable price regimes. We report this as sensitivity to temporal distribution shift rather than as a demonstrated price-level mechanism: blocking changes several things at once, including price-range overlap, regime composition, temporal distance between partitions, and whether training data may post-date test data. We have not isolated which of these drives the reversal. With blocking, all three assets show a positive microstructure gap, ranging +0.027 to +0.049 at the reported phase and positive across all ten phase rotations (Section 6.3).

Price features behave inconsistently and we do not rely on them. On ETH they move from −0.000 to +0.037 under blocking; on SOL from +0.031 to +0.012. We had expected blocking to reduce price-feature performance uniformly by removing a time proxy. It did not, and we have no account of the ETH reversal. Headline figures therefore use the microstructure subset, which excludes direct price-level columns and directional lagged-return features, and behaves consistently under both schemes.

6.3 Rotating the block phase

The block assignment uses one phase: blocks are assigned by index modulo 10, with 0–6 training, 7–8 validation and 9 test. That phase is arbitrary, so we ran all ten rotations, holding the 150 blocks, the 990-second embargo, the 63-feature representation and the model specification fixed. Phase p assigns roles by (block index $+\,p$) mod 10. The resulting test sets are disjoint across phases and together cover 95% of eligible snapshots, so each retained snapshot serves as test data in exactly one rotation. The training sets overlap heavily and all rotations draw on the same underlying series, so the rotations are not statistically independent.

Table 7: Microstructure gap over lookup across 10 block phases, 63-feature representation. Every rotation is positive on every asset.

Asset	Mean	SD	Min	Max	Positive
BTC/USD	+0.0473	0.0052	+0.0378	+0.0529	10/10
ETH/USD	+0.0382	0.0144	+0.0130	+0.0550	10/10
SOL/USD	+0.0346	0.0103	+0.0133	+0.0463	10/10

All 30 rotations are positive. The assignment used in Table 6 is phase 0 under this parameterisation, identifiable by its lookup values matching that table exactly on all three assets. It is included above and is not an outlier: it gives +0.0483, +0.0484 and +0.0273 against rotation means of +0.0473, +0.0382 and +0.0346. Those differ from Table 6's +0.0479, +0.0491 and +0.0274 by at most 0.0007, the rotations using one seed against three.

The standard deviations in Table 7 summarise dispersion across the ten block-phase assignments. They are not a general estimate of sampling or partition uncertainty, since the phases share training data and one underlying series. They exceed the seed standard deviations of Table 3 by roughly an order of magnitude, which is the expected ordering.

Two rotations sit close to zero: ETH/USD phase 7 at +0.0130 and SOL/USD phase 1 at +0.0133. The effect is consistently signed but small enough that an individual partition can nearly absorb it.

6.4 The gap is not consistently associated with price-range overlap

Price-range overlap varies across rotations, from 0.603 to 0.961 on BTC/USD, 0.712 to 0.977 on ETH/USD and 0.534 to 0.787 on SOL/USD. Since the blocked split was introduced to raise overlap, a natural concern is that the gap simply tracks it.

We observe no consistent association across the ten rotations. Correlations between overlap and gap are −0.17, +0.33 and −0.14, differing in sign by asset. Splitting each asset's rotations at its median overlap gives mean gaps of +0.0479, +0.0387 and +0.0373 in the low-overlap half against +0.0467, +0.0377 and +0.0320 in the high-overlap half: marginally larger where overlap is worse, on two of three assets.

We treated overlap as a measured covariate rather than a selection criterion. An earlier version of the rotation script required every rotation to exceed 0.7 overlap and aborted when BTC/USD (minimum 0.603) and SOL/USD (0.534) failed. Gating on overlap would have selected partitions by a property plausibly related to the outcome, which is the defect the rotation test exists to rule out.

6.5 The blocked split is exploratory

The blocked split was introduced after inspecting the temporal-split results, prompted by SOL/USD's negative gap, and is in that sense exploratory. Two things limit how much that concerns us. Block roles are assigned by time index and use no outcome information. And the result does not depend on the one arbitrary choice the scheme makes: all ten phase rotations are positive on all three assets (Section 6.3), so the reported partition is not a favourable selection from the available alternatives.

It remains an information-content diagnostic rather than a forecasting evaluation, for the reason given below.

6.6 Which scheme to believe

Neither replaces the other. The temporal split respects forecasting order and is the appropriate evaluation for a strategy question; the blocked split reduces temporal distribution shift between partitions, at the cost noted in Section 1.4. That the microstructure result holds under both on BTC and ETH is why we report it as a finding rather than a split-specific number.

7 Five Defects

Each of the following produced a plausible result. None was detectable by inspecting output. We list them with the check that caught each, because that is the transferable content.

1. Quote offset never reached the model. The conditioning vector carried only the latency window. For a fixed snapshot, label rows spanning four offsets presented identical model input with different targets, so the loss minimiser was to predict the mean and ignore the input. Symptom: validation loss moving 0.003 over 40 epochs. Effect: C-index 0.4749, rising to 0.6266 once corrected. *Caught by* a diagnostic asking whether a lookup table on the conditioning variables alone could beat the model. It could.

2. A synthetic 100 ms clock applied to real data. The label builder converted a latency window to a fixed snapshot count using an interval hardcoded for the synthetic generator. On real data, the "200 ms" label was a 33.5-second median horizon and "2000 ms" was 511.1 seconds, overstatements of 168× and 256×. *Caught by* an arithmetic fingerprint. That version used a $4 \times 4 \times 2 = 32$ label grid (four offsets, four latency windows, two sides), against the $4 \times 2 \times 2 = 16$ grid used in this paper. Reported label counts fell short of snapshots × 32 by exactly 672 rows on every asset, which is 21×32, and 21 is the tail a loop bounded at $n - 2000/100 - 1$ leaves behind. A horizon resolved against real timestamps would have left a different remainder per asset, since their gap distributions differ.

3. Microsecond timestamps read as nanoseconds. The replacement label builder converted timestamps with `astype("int64") / 1e9`, but Parquet returns `datetime64[us]`. Every interval came out 1000× too small, so a "30-second" window contained 920 snapshots and crossing rates reached 0.97 with no offset gradient. *Caught by* a printed diagnostic showing a median gap of 0.0 s against a known 11.5 s. The current builder raises if the median gap is below 1 second.

4. A survival head fed binary targets. When labels became binary, they were passed to a time-to-fill decoder with observed time set to a constant. Training loss sat at 16.1181 from the first epoch and test AUC was exactly 0.5000 on all six arms. *Caught by* a positive control: the same architecture with a binary head reaches AUC 0.9861 on synthetic data with a planted signal, so the failure was the head, not the data.

5. Split proportions computed on time rather than count. Cutting the time range at 70/85% gave partitions of 44/31/25% because snapshot density varies across the window. *Caught by* asserting the realised proportions.

7.1 What generalises

The common feature is that every defect produced output within the range a reader would accept. C-index of 0.5526 looks like a weak model. AUC of 0.8264 looks like a strong one. Neither was real.

Three practices follow. Run the trivial reference before the model, since four of the five defects would have been visible as a model failing to beat a lookup table. Write a positive control on planted signal, since defect 4 is undetectable any other way. Assert the properties of your splits rather than assuming them.

We also note the cost of misattributing a failure. After defect 1, we read a flat learning curve as a sample-size problem, projected that 300,000 snapshots would be required, and collected for six additional weeks. The projection was extrapolated from noise generated by the defect.

8 Limitations

No economic evaluation. AUC is not profit, and we do not model transaction costs, queue position or adverse selection.

The label is a crossing proxy, not an execution. Labels come from replaying book snapshots and use no trade data. A quote is recorded as crossed when the opposite best quote reaches or crosses its price. This is optimistic for a late queue arrival, since a crossing does not imply the resting order executed.

Brief crossings between snapshots are missed. With median inter-update gaps of 11.5 to 48.0 seconds, a crossing that occurs and reverses within a single gap leaves no trace in the data and is labelled as a non-crossing. The direction of the resulting bias is not established: it depends on whether such transient crossings correlate with the microstructure state at submission.

Two horizons. 300 and 900 seconds, chosen because observability at 30 and 60 seconds fell to 41–54% on ETH and SOL.

Observability selection. Labels exist only where at least one subsequent book snapshot falls inside the horizon. Because the feed is event-driven, update intensity is state-dependent, and quiet states are more likely to be unobservable. Dropping those rows may therefore select toward more active market states. At the worst case, ETH/USD at 300 s, 11.3% of potential observations are dropped. We do not quantify this selection effect, so our results are more precisely stated as predictive content for the crossing outcome *conditional on sufficient subsequent book observability.*

One venue, one window, three assets.

One deep architecture. An LSTM at one hyperparameter setting. The ETH LSTM-all arm is likely undertrained.

Blocked-phase sensitivity is quantified; general partition uncertainty is not. Seed standard deviations in Table 3 vary model initialisation only. Table 7 reports dispersion across the ten exhaustive phase rotations of this blocked scheme, which is not a general estimate of sampling or partition uncertainty: the rotations share training data and one underlying series. No equivalent distribution exists for the temporal split, which admits a single forecasting-ordered cut.

The price-feature behaviour is unexplained. We report it and exclude those features from headline results rather than offering an account we cannot support.

9 Conclusion

Given side, quote offset and holding horizon — all known to the trader for free — limit order book microstructure adds roughly 0.04 AUC to predicting our replay-defined crossing outcome on this data. Under a blocked split the gain is positive across three cryptocurrency pairs and all ten block-phase rotations, 30 of 30, with means of +0.047, +0.038 and +0.035. Under a forecasting-ordered temporal split it holds on two of three assets, and the third is sensitive to temporal distribution shift rather than showing absent signal.

That is a measured effect and a small one. It is smaller than the numbers we reported in three previous versions of this work, each of which was inflated by a defect in our own pipeline. We consider the catalogue in Section 7 at least as useful as the measurement, because the defects were not exotic and the outputs they produced were not obviously wrong.

The question of whether a 0.04 AUC edge over a lookup table is worth anything after costs is the natural next one, and this paper does not answer it.

Data and Code Availability

The collection pipeline, label builder, models, verification code and every diagnostic reported here are available at `https://github.com/azure-cyber/latency-compensating-mm`. The verification harness (`verify_v3.py`) runs the positive control and split assertions described in Section 7 and exits non-zero on failure.

Disclosure of AI Assistance

The author used Claude (Anthropic) extensively throughout this project: for implementation of the data collection pipeline, feature engineering, models and evaluation code; for diagnostic analysis; and for drafting and revision of this manuscript. Several of the defects catalogued in Section 7 originated in AI-generated code, and several were identified through AI-assisted diagnosis. The author designed the study, directed the analysis, made all substantive research decisions, verified the results, and is solely responsible for the content and any errors.

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